# Tasnim Jahan

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GitHub — Google Scholar — LinkedIn — Website

## Research Interests

• Computer Vision • Medical Image Analysis • Domain Adaptation • 2D/3D Image Segmentation • Representation Learning • Few-Shot Learning • Transfer Learning • Deep Learning

## Education

**M.Sc. in Computer Science and Engineering (Data Science), CGPA: 3.92 / 4.00** Oct 2021 – Jun 2024
United International University *Dhaka, Bangladesh*
*Summa Cum Laude (Highest Distinction), CGPA: 4.00 / 4.00 in last 5 semesters*

**M.Sc. in Mathematics, CGPA: 3.77 / 4.00** Jan 2016 – Apr 2017
Comilla University *Cumilla, Bangladesh*
*Thesis: Bernoulli's Equation with the Variation of Velocity, Density, and Diameter of Pipe; Advisor: Zillur Rahman*

**B.Sc. in Mathematics, CGPA: 3.61 / 4.00** Mar 2010 – Jan 2016
Comilla University *Cumilla, Bangladesh*
*Project: Numerical Study for Solving Initial Value Problems in Ordinary Differential Equations Using Single-Step and Multi-Step Methods; Advisor: Prof. Dr. Khalifa Mohammad Helal*

**Higher Secondary Certificate (HSC), Science, GPA: 5.00 / 5.00** Jul 2006 – Sep 2008
Mvi. Shamsul Karim College *Feni, Bangladesh*

**Secondary School Certificate (SSC), Science, GPA: 5.00 / 5.00** Jan 2004 – Jun 2006
Chhagalnaiya Pilot Girls' High School *Feni, Bangladesh*

## Research Experience

**Independent Researcher (Computer Vision & Deep Learning)** Jan 2024 – Present
*Remote*

*Advisors: Prof. Andrew King (King's College London, UK); Prof. Swakkhar Shatabda (BRAC University, Bangladesh)*

- Conducting research on few-shot learning, transfer learning, visual representation learning, and cross-domain adaptation using CNNs, Vision Transformers, and pretrained vision foundation models for robust visual learning.
- First author of a Q1 journal manuscript under revision on a lightweight attention-enhanced framework for ultrasound image segmentation.
- Co-author of an IEEE SMC 2025 paper on efficient deepfake detection using sparse subnetworks.

## Publications

- L. Al Amin, Md. Ismail Hossain, Thanh Thi Nguyen, **Tasnim Jahan**, Mahbubul Islam, Faisal Quader. **Uncovering Critical Features for Deepfake Detection through the Lottery Ticket Hypothesis**. *IEEE SMC*, 2025.
- **Tasnim Jahan**, Md Easin Arafat, Swakkhar Shatabda. **ThyCLIPNet: A BiomedCLIP-Guided Lightweight Attention-Enhanced DeepLabV3+ Framework for Robust Thyroid Nodule Segmentation**. Under revision (Q1 journal).
- **Tasnim Jahan**, Andrew King, Swakkhar Shatabda. **HyProDINO: A Hybrid Prototype-guided DINO Framework for Cross-Domain Few-Shot Medical Image Segmentation.** Accepted for oral presentation at the Medical Imaging with Machine Learning (MIML) Workshop, MICCAI 2026.

## Awards & Scholarships

**MICCAI 2026 RISE Travel Grant** — Recipient of competitive travel support to attend MICCAI 2026 Aug 2026

**Merit-based Scholarship (5 times)** at United International University for academic performance Jan 2023 – Aug 2024

**Ranked 5th** in LISA 2025 Challenge, RISE-MICCAI Summer School May 2025 – Oct 2025

**Mentee** in RISE-MICCAI Mentorship Program Oct 2025 – Oct 2026

## Projects

**3D Hippocampus Segmentation (Ultra-low-field MRI)** | *Python, MONAI, PyTorch* | Code Jul 2025 – Sep 2025

- Developed a MONAI-based 3D deep learning pipeline for bilateral hippocampus segmentation from ultra-low-field MRI.
- Implemented and compared Attention U-Net, DynUNet, SegResNet, and UNETR using class-balanced cropping and sliding-window inference.
- Among the tested models, Attention U-Net achieved the highest validation Dice ($\approx 0.60$) with post-processing.

**Thyroid Nodule Segmentation** | *Python, TensorFlow/Keras* | Code — Report Nov 2023 – Feb 2024

- Developed an end-to-end deep learning pipeline for thyroid nodule segmentation on the TN3K ultrasound dataset.
- Implemented and compared U-Net variants with pretrained encoders (VGG16, VGG19, ResNet50, MobileNetV2).
- Best among the tested encoders: MobileNetV2-based U-Net (Dice: 84.70%, IoU: 73.80%), achieving strong accuracy with a lightweight backbone.

**Spatio-Temporal Data Analysis Using Deep Learning (Survey Paper)** | *Literature Review* | ReportJul 2023 – Nov 2023

- Conducted a structured survey of deep learning methods for spatio-temporal data: data types, pipelines, and models.
- Covered CNNs, RNNs, GNNs, and Transformers for prediction, classification, clustering, and anomaly detection.
- Summarized real-world applications, open challenges, and research directions.

**Hypo-thyroid Disease Prediction** | *Python, Scikit-learn* | Code — Report Feb 2022 – May 2022

- Developed a machine learning pipeline for early detection of hypothyroid disorder using clinical and demographic data.
- Applied preprocessing techniques including imputation, feature encoding, scaling, and SMOTE for class imbalance.
- Trained and compared Decision Tree, Logistic Regression, SVM, and Random Forest classifiers.
- Achieved the best performance with Random Forest (Accuracy: 98.70%), outperforming other models.

**Face Mask Detection** | *Python, TensorFlow, OpenCV* | Code — Report Mar 2023 – Jul 2023

- Implemented a full pipeline for mask vs. no-mask detection using a MobileNetV2-based CNN.
- Included training, evaluation, and real-time inference with OpenCV.

**Predict Customer's Future Purchases** | *Python, Machine Learning* | Code — Report Jun 2022 – Sep 2022

- Developed a machine learning pipeline to predict customer purchases within 90 days using transactional retail data.
- Trained and evaluated Logistic Regression, Decision Tree, Random Forest, and Gaussian Naïve Bayes.
- Achieved the best performance with Logistic Regression (Accuracy: 99.10%, F1-score: 98.50%), while GaussianNB provided the fastest inference.

**Bangla Financial Sentiment Classification** | *Python, ML* | Code — Report Jan 2023 – Aug 2023

- Collected and labelled 1.8K+ Bangla financial headlines from the Prothom Alo newspaper for sentiment classification.
- Achieved the best performance with SVM (F1: 78%) and Logistic Regression (Accuracy: 76.60%).

**LLM-Enhanced Stock Movement Prediction** | *ML* | Report Aug 2023 – Sep 2023

- Integrated financial time-series data with information extracted from news using LLMs for weekly stock prediction.
- Improved Logistic Regression (Accuracy: 54.80% $\rightarrow$ 63%, F1: 42.10% $\rightarrow$ 60.90%).

**IoT-Based Watering Automation and Security System for Plants & Pets** | *Sensors* | Report Jul 2022 – Nov 2022

- Designed an IoT concept for automated watering with basic security monitoring for plants and pets.
- Focused on automation logic, sensor-driven triggers, and system-level design.

## Skills

**Programming**: Python, SQL, Java, C, C++
**Deep Learning**: PyTorch, TensorFlow, Keras, MONAI
**Computer Vision / ML**: OpenCV, scikit-learn, NumPy, Pandas
**Research Tools**: LaTeX, Kaggle, Google Colab, Jupyter Notebook
**Visualization**: Matplotlib, Power BI
**Soft Skills**: Complex problem solving, rapid skill acquisition, effective communication

## Relevant Coursework

- Advanced Artificial Intelligence
- Deep Learning
- Machine Learning
- Data Mining
- Natural Language Processing
- Research Methodology
- Advanced Database Systems
- Differential & Integral Equations
- Discrete Mathematics
- Linear Algebra
- Numerical Analysis
- Ordinary & Partial Differential Equations
- Methods of Statistics
- Vector Calculus & Tensor Analysis

## Teaching Experience

**Section Leader (Volunteer)** — Apr 2026 – May 2026
*Stanford University – Code in Place* — *Remote*
- Selected through a competitive process from 2,000 applicants to join a global teaching team.
- Taught Python programming to a diverse cohort in a large-scale online course.
- Led problem-solving sessions and mentored students to improve programming and analytical skills.

**Instructor – Mathematics** — Jan 2017 – Mar 2018
*Institute of Computer Science and Technology (ICST)* — *Feni, Bangladesh*
- Taught mathematics to 200+ diploma-level students across engineering disciplines.
- Designed lectures, assignments, and assessments tailored to diverse learning needs.

**Junior Teacher – Mathematics** — Mar 2018 – Dec 2018
*Mainamati English School and College* — *Cumilla, Bangladesh*
- Delivered mathematics instruction in English medium and monitored student performance.
- Founded and led a math club to improve student engagement.

## Professional Experience

**Freelance Data Analyst** — Jan 2023 – Feb 2024
*Fiverr* — *Remote*
- Developed interactive dashboards and data-driven reports using Microsoft Power BI.
- Performed data cleaning, statistical analysis, and feature engineering on real-world datasets.

**Field Data Collector** — Nov 2021 – Dec 2021
*BRAC James P Grant School of Public Health* — *Dhaka, Bangladesh*
- Collected qualitative and quantitative data from ∼100 participants using Kobo Toolbox.
- Ensured data quality, consistency, and proper documentation for analysis.

**Programme Organizer – Education** — Jan 2022 – Apr 2022
*Jagorani Chakra Foundation* — *Cox's Bazar, Bangladesh*
- Supported the implementation of a UNICEF-funded education program by training teachers in effective teaching methodologies and classroom practices.
- Collaborated with stakeholders, monitored field activities, and contributed to reporting and documentation.

**Education Facilitator – Education** — Jan 2019 – Apr 2020
*DanChurch Aid* — *Cox's Bazar, Bangladesh*
- Delivered literacy and numeracy sessions to ∼300 learners in a UNICEF-funded program.
- Designed course materials and conducted baseline and endline data collection.

## Online Courses & Certifications

**WorldQuant University**: Computer Vision Lab — Dec 2025 – Present

**RISE-MICCAI Summer School** (MICCAI Society) — Jul 2025

**Stanford University**: Code in Place — Apr 2023 – Jun 2023

**Google**: Data Analytics Specialization — Apr 2021 – Jun 2021

**University of Amsterdam**: Basic Statistics — Jun 2021 – Jul 2021

## Academic Service

- **Journal Reviewer**, Biomedical Signal Processing and Control (Q1) — reviewed 6 manuscripts.
- **Workshop Reviewer**, MICCAI 2026 MSB EMERGE and FAIMI–BRIDGE–EPIMI — reviewed 3 manuscripts.

## Extracurricular Activities

**English & Bangla News Presenter** at Bangladesh Betar, Cox's Bazar, Bangladesh — Apr 2025 – Present

**Member** at Partibartan, Cultural Organization, Comilla University, Cumilla, Bangladesh — Jan 2014 – Jan 2017